

3 (a) angular speed of grating, $\omega = 2 \pi / T = 2 \pi / 3.0 = 2.1 \text{ rad s}^{-1}$ A1

(b) The peaks represent the positions of constructive interference/maxima. B1

B1

The effect of diffraction through the slits of diffraction grating causes interference fringes of higher order to have lower intensity than the zeroth order maxima.

(Peak C corresponds to the zeroth order of the diffraction pattern whereas peaks B and D the first order and Peaks A and E the second order.)

(c) (i) Time interval = $3.7 \times 0.1 = 0.37 \text{ s}$ C1

$\theta = \omega t = 2.1 \times 0.37 = 0.78 \text{ rad}$ (ecf) A1

(c)(ii) Using the grating equation $n \lambda = d \sin \theta$, $d = 1 \times 10^{-3} / 550$ C1

$$\sin \theta_2 = 2 \times 5.5 \times 10^5 \times \lambda$$

$\lambda = 640 \text{ nm}$ (ecf) A1

(c)(iii) 1 Peak E is preferred as the angle θ is larger so the percentage uncertainty for calculating the wavelength is smaller. C1

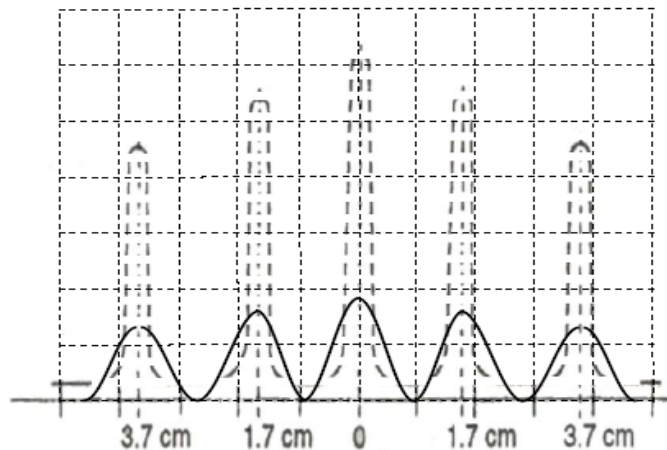
(c)(iii)2 From $n \lambda = d \sin \theta$

$$n < d/\lambda = 1 \times 10^{-3} / 550 \times 640 \times 10^{-9} = 2.8 \quad \text{C1}$$

Hence, the highest order observed is 2nd order maxima. C1

$$\text{Total number of orders observed} = 2n + 1 = 5 \quad \text{A0}$$

(d) Diagram must show that since the slit separation remains the same, the fringe separation remains the same. Peaks are less intense (poorer contrast) and less sharp (broader and less defined). M1



4 (a) (i)

2 25/0 0108 = 208 0

C1